

Mohamed Amara

3B Mechatronics Engineering | Available for co-op: Winter 2027 onward
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Education

University of Waterloo

Expected April 2028

Candidate for Bachelor of Applied Science, Honours Mechatronics Engineering

Relevant coursework: Sensors and Instrumentation, Circuits, Digital Logic, Solid Mechanics, Computer Structures and Real-Time Systems, Engineering Design

Technical Skills

Languages: C++, Python, TypeScript, JavaScript, Java, MATLAB

Embedded & lab: STM32, sensors, UART, SPI, I²C, oscilloscope, logic analyzer, soldering

Engineering: SolidWorks, AutoCAD, KiCad, UML, hardware feasibility, requirements analysis

Software: Node.js, React, Flutter/Dart, MongoDB, Git

Experience

AI/Hardware Systems Developer | The Plug Co

Fall 2025 & Summer 2026

Memory Box: AI software for organizing and searching photos, audio and video

- Collaborated on Python AI pipelines for processing camera, audio and motion data into structured events for searchable memories.
- Contributed to TypeScript device integration, including data validation, timestamp checks and recovery after disconnects.
- Created the initial UML deployment diagram to map connections between client devices, backend services and storage.
- Compared cloud and on-device AI in a hardware feasibility study; recommended a phone-based first release to avoid requiring custom hardware.

Software Engineer Intern | CoinWa

Jan 2025 - Apr 2025

Remote

- Developed Flutter/Dart screens for interactive budgeting and saving lessons; resolved rendering and performance issues with Flutter DevTools.
- Built Node.js and MongoDB services for user accounts and learning progress, including email verification and password recovery.

Software Engineering Intern | Western University

Feb 2023 - Jun 2023

London, Ontario

- Developed a Java tool to assign instructors to courses using qualifications and availability, improving allocation efficiency by approximately 80%.
- Refined requirements with departmental staff, trained users and documented the application for ongoing use.

Selected Projects

Human-Powered Washing Machine

SolidWorks, mechanical design

- Modelled the drum, frame, seat and pedal drivetrain; designed two chain-drive ratios for washing and spin drying.
- Calculated shaft loads, torque and fatigue to size transmission components for a target safety factor of 2.

OEM CarPlay Integration - 2016 Acura ILX

In development: research & design

- Outlined an SBC/STM32 design to add wireless CarPlay while preserving the factory displays, controls and backup camera.
- Planning donor-unit signal measurements and fail-safe video switching, with factory-system fallback and camera priority as core requirements.